

CODAPath: Contrastive Optimization via Dual-Foundation Models for Active Learning in Pathology

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Abstract

Training high-performance pathology models demands large-scale annotated datasets, requiring significant financial and temporal costs. Active Learning (AL) offers a solution by annotating only the most informative samples, thereby reducing the labeling effort. Nevertheless, most AL methods drop in performance under extreme low-budget regimes and frequently rely on naive random selection during the initial query phase. In computational pathology, where images exhibit profound morphological complexity and redundancy, this forced “cold-start” suffers from severe selection bias and wastes the critical initial budget. To overcome this, we propose CODAPath, a novel AL framework that completely eliminates random initialization by leveraging Medical Vision-Language Models (Med-VLMs) as zero-shot priors. CODAPath employs a joint greedy acquisition strategy that concurrently evaluates pseudo-prediction uncertainty and spatial coverage, assembling a highly diverse initial batch from step zero. Furthermore, we improve fine-tuning under limited supervision by integrating Center Loss to enforce highly separable class representations. Notably, with a budget of just 50 annotated samples (0.2%) on the HistoSet-5x14 dataset, CODAPath achieves an accuracy of 79.25%, outperforming random sampling by 15.32% and the second-best baseline (Margin) by 9.84%. Further experiments on three public pathology image datasets show that CODAPath establishes a strong baseline for extreme low-budget active learning in medical imaging. The code is available at <https://github.com/CryAndRRich/codapath>.

Keywords: active learning, low-budget learning, pathology image classification, vision-language models.

1. Introduction

Pathology serves as the gold standard for clinical diagnosis. The digitization of whole-slide images has catalyzed a paradigm shift toward computational pathology, where high-capacity deep learning architectures, such as Convolutional Neural Networks and Vision Transformers, achieve remarkable success in automated tissue phenotyping and disease grading [1]. However, optimizing these models depends heavily on large-scale, accurately annotated datasets, which are prohibitively expensive and time-consuming to acquire from board-certified pathologists. To alleviate this annotation bottleneck, Active Learning (AL) is widely adopted to iteratively query an oracle for only the most informative data subset [2].

While effective under moderate budgets, contemporary AL frameworks degrade in extreme low-budget scenarios due to the “cold-start” problem: an initially untrained model should rely on naive random sampling to construct its first training batch [3]. In histopathology, this reliance is particularly inefficient. Because pathological images exhibit profound visual complexity and vast regions of redundant tissue, random sampling under tight constraints suffers from severe selection bias, wasting a crucial portion of the initial labeling budget. Furthermore, standard AL methods typically fail to concurrently evaluate uncertainty and coverage. An optimal extreme low-budget strategy

should target samples that are both difficult to classify and structurally varied enough to represent the entire data distribution.

To overcome the cold-start problem without incurring additional annotation costs, domain-specific Medical Vision-Language Models (Med-VLMs), which are pre-trained on vast clinical corpora, offer a promising avenue by providing informative semantic priors tailored for histopathology [4, 5]. Building upon this insight, we propose CODAPath, a novel Active Learning framework designed to tackle the extreme low-budget regime. Rather than relying on blind heuristics, CODAPath leverages Med-VLM text-image alignments to estimate sample informativeness before any explicit labeling occurs. Furthermore, it incorporates an auxiliary metric learning objective during the training process to stabilize the model against severely limited annotations.

The specific contributions of this work are summarized as follows:

- We introduce a novel acquisition objective that concurrently evaluates predictive uncertainty and spatial coverage to select informative initial data points. Crucially, this strategy is powered by robust, unbiased semantic priors extracted from a dual-foundation Med-VLM architecture, eliminating the reliance on random initialization.

- To overcome representation collapse under extreme data scarcity, we integrate Center Loss [6] during the fine-tuning phase, enforcing highly separable class features even with a minimally constrained dataset.
- Extensive experiments demonstrate that CODAPath outperforms existing AL baselines under extreme low-budget constraints across three public pathology datasets.

2. Related Work

Uncertainty and Coverage in Active Learning. Traditional AL approaches mitigate the annotation bottleneck by prioritizing samples based on two primary heuristics: uncertainty and coverage [7–9]. Uncertainty-based methods greedily select instances for which the current model produces the least confident predictions, utilizing strategies such as Entropy, Margin sampling, Bayesian AL (BALD), and Ensemble-based disagreement [10–13]. Conversely, coverage-based methods aim to select a representative subset that geometrically covers the entire data distribution, employing techniques like CoreSet, Variational Adversarial AL (VAAL), and Learning Loss [14–16]. To leverage the strengths of both paradigms, hybrid approaches such as BADGE [17], ALFA-Mix [18], and WAAL [19] attempt to balance uncertainty and coverage by sampling high-magnitude points within hallucinated gradients or feature spaces. However, these frameworks inherently assume the availability of a pre-existing, well-calibrated labeled pool to compute meaningful probabilities and embeddings.

The Cold-Start Problem. In extreme low-budget regimes, the assumption of an initial labeled pool completely breaks down. An untrained model produces highly uncalibrated, noisy probabilities, rendering early uncertainty estimates practically uninformative. Consequently, traditional and hybrid methods are forced to rely on naive random sampling for initialization, known as the cold-start problem [20, 21]. To explicitly address this data-scarce environment, recent AL frameworks have shifted focus toward initial pool construction. TypiClust [3] bypasses random initialization by clustering dense self-supervised features and querying the most typical examples from high-density regions. Similarly, ActiveFT [22] optimizes a continuous parametric model to preserve the overall data distribution. Furthermore, recent advancements have formulated sample selection as sophisticated facility location and generalized coverage problems, as seen in ProbCover [23], MaxHerding [24], DCoM [25], and Uncertainty Herding [26].

Vision-Language Models in Active Learning. Although specialized cold-start methods successfully mitigate random initialization, they rely strictly on unimodal visual features. In computational pathology, substantial morphological variation and staining artifacts

can limit the usefulness of unsupervised visual representations without task-specific guidance. To address this without additional annotation cost, recent work has introduced Vision-Language Models (VLMs) into active learning. For instance, frameworks like CEC [27] leverage general-domain VLM models (e.g., CLIP [28]) to guide acquisition. However, models trained on natural images may transfer less effectively to pathology under extreme low-budget settings. Other approaches, such as PCB [29], formulate active learning specifically for prompt tuning to optimize the VLM, which is a fundamentally different objective than efficiently acquiring a labeled pool to train a downstream task model. To overcome these limitations, our CODAPath framework shifts the paradigm by integrating medical-specific multimodal models (Med-VLMs), such as PLIP [4] and BiomedCLIP [5], which are pre-trained on vast clinical corpora. Instead of relying on general-domain priors or optimizing the foundation model itself, CODAPath leverages Med-VLM zero-shot pseudo-probabilities to concurrently evaluate predictive uncertainty and feature-space diversity, establishing a highly informative initial pool from step zero.

3. Methodology

In a standard pool-based Active Learning (AL) setup, a model f_θ is iteratively trained by querying a budget B of informative samples from a large unlabeled pool $\mathcal{U}$ to populate a labeled set $\mathcal{L}$. However, in the extreme low-budget and cold-start scenario where the initial labeled pool is strictly empty ($\mathcal{L}^{(0)} = \emptyset$), conventional AL fails due to the lack of model weights to compute gradients or probabilities. CODAPath bypasses this reliance on random initialization by formulating a direct zero-shot acquisition strategy at step $t = 0$. The detailed architecture and workflow of our framework are illustrated in Fig. 1. The pipeline consists of three core components: (1) Zero-Shot Priors via Dual-Foundation Med-VLMs; (2) Sampling via Uncertainty and Coverage; and (3) Contrastive Optimization via Center Loss. The overall step-by-step procedure is formally summarized in Algorithm 1.

3.1. Zero-Shot Priors via Dual-Foundation Med-VLMs

To establish a robust decision boundary without task-specific weights, CODAPath leverages the zero-shot capabilities of Medical Vision-Language Models (Med-VLMs). However, relying on a single foundation model risks introducing pre-training distributional biases, as models differ in their focus. To mitigate this and maximize representational ability, we propose a dual-foundation architecture that ensembles two complementary Med-VLMs: PLIP [4] (highly specialized in pathology morphology) and BiomedCLIP [5] (robust across broader biomedical contexts). This dual-VLM strategy ensures a more comprehensive image-text alignment, providing stable and unbiased zero-shot decision boundaries to guide the subsequent uncertainty and coverage sampling.

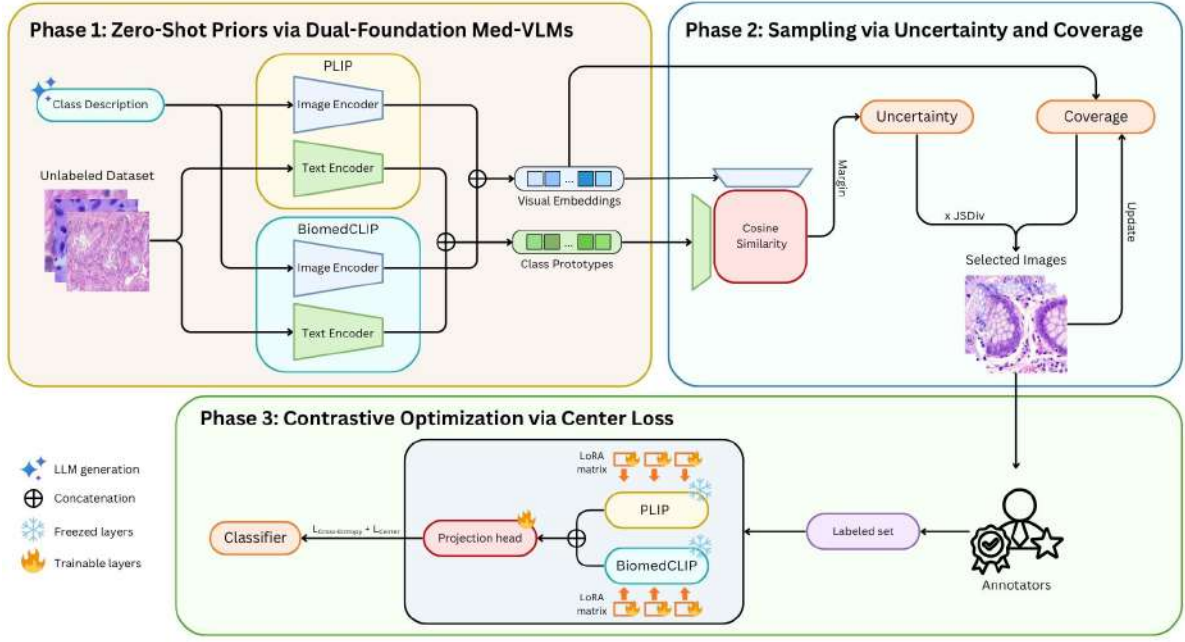


Fig. 1. An illustration of our CODAPath framework.

Rather than relying on raw class names, we utilize an LLM to generate detailed, clinically accurate morphological descriptions d_c for each of the C classes (see Appendix A.1 for representative examples). To stabilize text representations, d_c is injected into K domain-specific prompt templates ($K = 5$, detailed in Appendix A.2), yielding $\{p_{c,k}\}_{k=1}^K$. Each prompt is independently encoded by PLIP (E_{PLIP}) and BiomedCLIP (E_{Bio}), and concatenated to form a unified semantic feature:

$$\mathbf{e}_{c,k} = [E_{\text{PLIP}}(p_{c,k}) \oplus E_{\text{Bio}}(p_{c,k})] \quad (1)$$

The final text prototype $\hat{\mathbf{t}}_c$ for class c is obtained by averaging across the K templates followed by L_2 normalization:

$$\mathbf{t}_c = \frac{1}{K} \sum_{k=1}^K \mathbf{e}_{c,k}, \quad \hat{\mathbf{t}}_c = \frac{\mathbf{t}_c}{\|\mathbf{t}_c\|_2} \quad (2)$$

Concurrently, for each unlabeled image $\mathbf{x}_i \in \mathcal{U}$, we extract, concatenate, and L_2 -normalize its visual features via the dual-foundation image encoders to obtain $\hat{\mathbf{v}}_i$. The zero-shot pseudo-probability distribution $P_i \in \mathbb{R}^C$ is then computed via a temperature-scaled Softmax function over the cosine similarities [30]:

$$P_{i,c} = \frac{\exp(\hat{\mathbf{v}}_i^\top \hat{\mathbf{t}}_c / \tau)}{\sum_{j=1}^C \exp(\hat{\mathbf{v}}_i^\top \hat{\mathbf{t}}_j / \tau)} \quad (3)$$

where $\tau = 0.05$ is the temperature hyperparameter. This distribution P_i serves as the foundational prior to evaluate sample informativeness.

3.2. Sampling via Uncertainty and Coverage

Using P_i , CODAPath constructs a joint acquisition strategy balancing predictive uncertainty and global feature-space coverage.

Algorithm 1 Our proposed CODAPath algorithm

Require: Unlabeled pool $\mathcal{U}$, budget B , categories C , dual image/text encoders, trade-off α .

- 1: **% Phase 1: Zero-Shot Priors via Dual-Foundation Med-VLMs**
- 2: **for** each class $c \in C$ **do**
- 3: Generate templates and compute unified text prototype $\hat{\mathbf{t}}_c$ via Eq. 2.
- 4: **end for**
- 5: **for** each sample $\mathbf{x}_i \in \mathcal{U}$ **do**
- 6: Extract visual feature $\hat{\mathbf{v}}_i$ and compute pseudo-probability P_i via Eq. 3.
- 7: Compute and normalize prediction uncertainty $\tilde{U}_i$ via Eq. 4–6.
- 8: **end for**
- 9: **% Phase 2: Sampling via Uncertainty and Coverage**
- 10: Initialize selected set $S \leftarrow \emptyset$.
- 11: **while** $|S| < B$ **do**
- 12: **for** each candidate $\mathbf{x}_k \in \mathcal{U} \setminus S$ **do**
- 13: Compute and normalize spatial coverage gain $\tilde{G}(\mathbf{x}_k|S)$ via Eq. 7.
- 14: **end for**
- 15: Select $\mathbf{x}^*$ via Eq. 8.
- 16: $S \leftarrow S \cup \{\mathbf{x}^*\}$.
- 17: **end while**
- 18: Update datasets: $\mathcal{L} \leftarrow S$, $\mathcal{U} \leftarrow \mathcal{U} \setminus S$.
- 19: **% Phase 3: Contrastive Optimization via Center Loss**
- 20: Initialize dual-backbone with LoRA modules.
- 21: Fine-tune model on $\mathcal{L}$ using $L_{\text{CE}} + \lambda L_{\text{center}}$ via Eq. 9.
- 22: **return** Optimized model f_θ .

Uncertainty Estimation. We first evaluate the model’s confusion between its top predictions. We quantify this using the prediction margin m_i between the highest ($P_{i,(1)}$) and second-highest ($P_{i,(2)}$) probabilities. A smaller margin indicates that the zero-shot prior is highly uncertain about the true class:

$$m_i = P_{i,(1)} - P_{i,(2)} \quad (4)$$

However, the margin only considers the top two classes and may fail to distinguish between probability distributions that are completely flat versus those with a clear secondary peak. To penalize flat and uninformative predictions across all classes, we incorporate the Jensen-Shannon Divergence (JSD), denoted as J_i , between P_i and its one-hot hard-label projection Q_i :

$$J_i = \frac{1}{2} \sum_{c=1}^C \left[P_{i,c} \ln \left(\frac{P_{i,c}}{M_{i,c}} \right) + Q_{i,c} \ln \left(\frac{Q_{i,c}}{M_{i,c}} \right) \right] \quad (5)$$

where $M_i = 0.5(P_i + Q_i)$. The combined uncertainty U_i is formulated by inversely weighting the margin and scaling by JSD:

$$U_i = (1 - m_i)(1 + J_i) \quad (6)$$

This metric is then normalized across the dataset to obtain $\tilde{U}_i \in [0, 1]$.

Spatial Coverage. Relying solely on uncertainty often leads to sampling redundant images from the same dense and confusing feature regions. To ensure the queried batch is morphologically representative of the entire dataset, we model coverage via a submodular facility location function [14, 23]. Let S denote the set of currently selected images. The marginal coverage gain of adding a candidate $\mathbf{x}_k \notin S$ is:

$$G(\mathbf{x}_k|S) = \sum_{i=1}^{|\mathcal{U}|} \max \left[0, s_{i,k} - \max_{\mathbf{x}_j \in S} s_{i,j} \right] \quad (7)$$

where $s_{i,j} = \hat{\mathbf{v}}_i^\top \hat{\mathbf{v}}_j$. The gain is dynamically normalized at each step to obtain $\tilde{G}(\mathbf{x}_k|S) \in [0, 1]$.

CODAPath iteratively selects the next sample $\mathbf{x}^*$ by greedily maximizing the joint objective:

$$\mathbf{x}^* = \arg \max_{\mathbf{x}_k \in \mathcal{U} \setminus S} [(1 - \alpha) \cdot \tilde{G}(\mathbf{x}_k|S) + \alpha \cdot \tilde{U}_k] \quad (8)$$

where α controls the exploration-exploitation trade-off. This repeats until the budget limit is reached, i.e., $|S| = B$.

3.3. Contrastive Optimization via Center Loss

Once the initial set $\mathcal{L} = \{(\mathbf{x}_i, y_i)\}_{i=1}^B$ is assembled, CODAPath fine-tunes the dual-backbone model. To mitigate overfitting, we apply Low-Rank Adaptation (LoRA) [31] to the attention modules. The fused features are mapped through a shared projection head to a 256-dimensional embedding $\mathbf{f}_{\text{proj}}$.

Because standard cross-entropy loss fails to strictly structure the underlying feature space geometry under severe data scarcity, CODAPath jointly minimizes both class-weighted cross-entropy (L_{CE}) and Center Loss (L_{center}) [6]. By pulling same-class embeddings toward a shared prototype, Center Loss explicitly enforces compact intra-class clusters and well-separated inter-class margins, maximizing the utility of the minimal labeled batch.

$$L = L_{\text{CE}} + \lambda \cdot L_{\text{center}}, \quad L_{\text{center}} = \frac{1}{2N} \sum_{i=1}^N \|\hat{\mathbf{f}}_{\text{proj},i} - \mathbf{c}_{y_i}\|_2^2 \quad (9)$$

where $\mathbf{c}_{y_i} \in \mathbb{R}^{256}$ is the learnable prototype for class y_i , N is the batch size, and λ is a weighting scalar.

4. Experiments

We evaluate CODAPath against a diverse set of state-of-the-art active learning (AL) baselines under extreme low-budget constraints, focusing on the strict cold-start scenario where no pre-trained weights are available.

Datasets. We benchmark all methods on three highly diverse public pathology datasets, representing varying levels of morphological complexity and diagnostic challenges (dataset partitioning is explicitly detailed in Appendix B.1): PathMNIST [32] (9 distinct colorectal tissue types), HistoSet-5x14 [33] (14 fine-grained classes across multiple organ systems requiring subtle benign/malignant distinction), and SkinTissue [34] (16 diverse dermatopathology classes including tumors and normal structures).

Active learning methods. We compare our proposed CODAPath with seven representative baselines categorized as follows: random selection; uncertainty-based methods (Margin [11], Entropy [10]); coverage-based methods (CoreSet [14]); hybrid methods balancing uncertainty and coverage (BADGE [17]); and modern cold-start methods that rely on unsupervised distribution matching (TypiClust [3], ActiveFT [22]).

We note that recent methods such as CEC [27] also explore the use of general-domain VLMs for active learning. However, we do not include CEC in our comparison because official source code is unavailable, and reproducing the method would require several implementation choices that may substantially affect the model’s true performance. In addition, CEC was developed for natural-image benchmarks, whereas our study focuses on pathology classification with medical-domain encoders. Replacing CEC’s backbone with Med-VLMs would introduce methodological changes beyond the original formulation. As a result, we focus on baselines with publicly described procedures that can be evaluated consistently under our shared training protocol.

Table 1. Top-1 accuracy (mean $\pm$ std) averaged over 5 independent trials across 3 datasets and 6 budget regimes. The **first**, **second**, and **third** best results for each setting are highlighted.

Dataset	Budget	Random	Margin	Entropy	CoreSet	BADGE	TypiClust	ActiveFT	CODAPath
PathMNIST	50	89.00 $\pm$ 0.01	88.54 $\pm$ 0.43	54.71 $\pm$ 0.01	80.61 $\pm$ 0.04	88.94 $\pm$ 0.02	89.29$\pm$0.03	89.37$\pm$0.03	92.34$\pm$0.50
	100	93.59$\pm$0.13	92.23 $\pm$ 0.56	87.44 $\pm$ 1.35	89.44 $\pm$ 0.09	92.63$\pm$0.32	89.21 $\pm$ 0.09	91.75 $\pm$ 0.01	92.70$\pm$0.23
	150	94.92$\pm$0.01	93.84 $\pm$ 0.36	84.76 $\pm$ 2.69	92.84 $\pm$ 0.06	95.11$\pm$0.06	93.57 $\pm$ 0.02	94.11$\pm$0.01	93.58 $\pm$ 0.07
	200	94.94$\pm$0.02	94.29 $\pm$ 0.40	92.91 $\pm$ 0.38	93.38 $\pm$ 0.02	95.32$\pm$0.01	94.44 $\pm$ 0.04	95.58$\pm$0.01	94.05 $\pm$ 0.25
	250	95.13$\pm$0.01	92.87 $\pm$ 0.67	93.15 $\pm$ 1.09	94.74 $\pm$ 0.02	95.06$\pm$0.19	94.23 $\pm$ 0.03	95.71$\pm$0.01	94.40 $\pm$ 0.44
	300	95.42$\pm$0.04	94.64 $\pm$ 0.59	92.76 $\pm$ 0.38	94.07 $\pm$ 0.01	95.17$\pm$0.01	94.89 $\pm$ 0.01	95.28$\pm$0.03	94.67 $\pm$ 0.70
HistoSet-5x14	50	63.93 $\pm$ 0.02	67.23 $\pm$ 2.02	49.05 $\pm$ 0.45	56.68 $\pm$ 0.02	65.59 $\pm$ 0.12	69.41$\pm$0.50	67.73$\pm$0.07	79.25$\pm$0.49
	100	77.75 $\pm$ 0.16	76.50 $\pm$ 1.25	58.27 $\pm$ 2.57	72.25 $\pm$ 0.04	78.86$\pm$0.05	79.41$\pm$0.20	76.75 $\pm$ 0.13	82.13$\pm$0.29
	150	82.36 $\pm$ 0.05	86.02$\pm$0.23	69.55 $\pm$ 0.35	77.46 $\pm$ 0.08	84.36 $\pm$ 0.05	85.71$\pm$0.04	82.61 $\pm$ 0.03	85.52$\pm$0.34
	200	86.84$\pm$0.02	88.66$\pm$0.32	75.00 $\pm$ 0.58	80.38 $\pm$ 0.01	86.52 $\pm$ 0.03	85.34 $\pm$ 0.02	86.23 $\pm$ 0.09	86.93$\pm$0.02
	250	87.98$\pm$0.01	90.36$\pm$0.09	73.54 $\pm$ 2.30	81.86 $\pm$ 0.02	87.61 $\pm$ 0.40	86.16 $\pm$ 0.02	87.64 $\pm$ 0.02	89.20$\pm$0.34
	300	88.64 $\pm$ 0.02	90.29$\pm$0.38	80.57 $\pm$ 0.36	83.88 $\pm$ 0.05	87.68 $\pm$ 0.14	88.93$\pm$0.07	88.20 $\pm$ 0.01	90.29$\pm$0.04
SkinTissue	50	73.01$\pm$0.03	70.53 $\pm$ 0.71	46.52 $\pm$ 2.19	63.35 $\pm$ 0.01	68.67 $\pm$ 0.01	73.73$\pm$0.01	72.32 $\pm$ 0.01	80.01$\pm$0.01
	100	77.68 $\pm$ 0.01	77.94 $\pm$ 0.45	58.63 $\pm$ 0.50	71.80 $\pm$ 0.13	75.04 $\pm$ 0.46	79.47$\pm$0.03	79.35$\pm$0.12	80.84$\pm$0.06
	150	83.20 $\pm$ 0.01	84.42$\pm$0.17	71.98 $\pm$ 0.25	76.26 $\pm$ 0.05	83.85 $\pm$ 0.26	83.71 $\pm$ 0.01	84.07$\pm$0.03	84.65$\pm$0.01
	200	84.68 $\pm$ 0.06	85.67$\pm$0.41	80.51 $\pm$ 0.46	78.15 $\pm$ 0.03	86.34$\pm$0.03	85.14 $\pm$ 0.10	85.02 $\pm$ 0.05	86.36$\pm$0.03
	250	86.11 $\pm$ 0.01	87.37$\pm$0.53	82.00 $\pm$ 0.48	77.48 $\pm$ 0.04	86.71$\pm$0.16	86.50 $\pm$ 0.01	86.38 $\pm$ 0.05	87.65$\pm$0.03
	300	87.17 $\pm$ 0.01	88.08$\pm$0.34	86.16 $\pm$ 0.02	78.29 $\pm$ 0.03	87.39 $\pm$ 0.01	87.60$\pm$0.09	87.36 $\pm$ 0.04	87.77$\pm$0.01

Random Uniformly select a random set S of B data points from the unlabeled pool $\mathcal{U}$.

Margin [11] Iteratively select $\mathbf{x}^* = \arg \min_{\tilde{\mathbf{x}} \in \mathcal{U}} P_1(y|\tilde{\mathbf{x}}) - P_2(y|\tilde{\mathbf{x}})$, where P_1 and P_2 are the highest and second-highest predicted probabilities.

Entropy [10] Iteratively select $\mathbf{x}^* = \arg \max_{\tilde{\mathbf{x}} \in \mathcal{U}} H(y|\tilde{\mathbf{x}})$, where $H(\cdot)$ is the Shannon entropy.

BADGE [17] Select points by applying the K -Means++ initialization algorithm on the hallucinated gradient embeddings of the loss with respect to the last layer weights.

Coreset [14] Iteratively select points using the k -Center-Greedy algorithm to minimize the maximum distance between unlabeled points and their nearest labeled neighbor: $\mathbf{x}^* = \arg \max_{\tilde{\mathbf{x}} \in \mathcal{U}} \min_{\mathbf{x}' \in \mathcal{L}} \|\tilde{\mathbf{x}} - \mathbf{x}'\|_2$.

TypiClust [3] Run a clustering algorithm (e.g., k -means) on dense self-supervised features. For each cluster, select a point with the highest “typicality” (local density) using m -Nearest Neighbors.

ActiveFT [22] Select a subset S that minimizes the continuous parametric distribution divergence between the selected set and the entire unlabeled pool: $\min_S D_{\text{KL}}(P(\mathcal{U})\|P(S))$.

Implementation. To ensure a fair comparison and isolate the effectiveness of the acquisition strategies, all evaluated methods share the identical dual-encoder LoRA backbone and optimization hyperparameters (detailed in Appendix B.2). Crucially, traditional uncertainty and hybrid methods (Margin, Entropy, and

BADGE) require an initialized model to compute gradients and probabilities. To accommodate this in the cold-start regime ($t = 0$), we evaluate these iterative methods over 5 consecutive query rounds (each consuming 20% of the budget). As the initial step lacks model weights, their first query round is strictly forced to use uniform random sampling to bootstrap the model. Subsequent rounds then utilize their respective acquisition logic. In contrast, zero-shot and purely unsupervised methods (our CODAPath, TypiClust, ActiveFT, Coreset) operate directly without this random bootstrap, allowing us to evaluate their ability to construct a high-quality initial labeled pool $\mathcal{L}$ from scratch.

Metrics. To evaluate the effectiveness of the acquisition strategies, we utilize top-1 classification accuracy on the held-out test set as our primary evaluation metric. To account for the stochastic variance introduced by the initial random bootstrapping and model optimization, all experiments are conducted across five independent trials using different random seeds. Consequently, we report the mean accuracy and its corresponding standard deviation ($\pm$ std) aggregated across these five trials.

4.1. Comparison with existing methods

The quantitative results across the three datasets are summarized in Table 1, with the mean accuracy differences relative to random sampling visualized in Fig. 2. Our analysis focuses on the model’s behavior

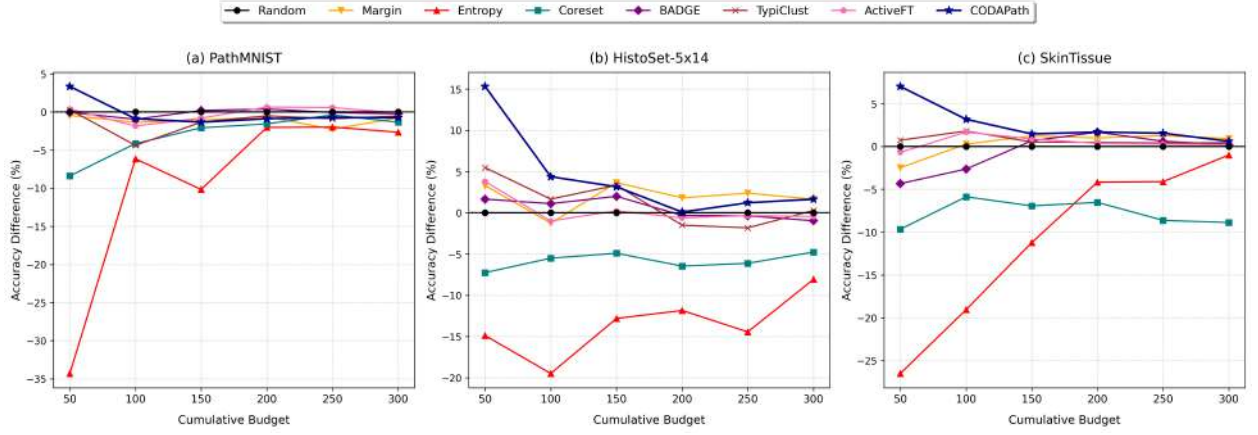


Fig. 2. Mean accuracy difference between several AL algorithms and random queries. Positive mean difference indicates that the corresponding AL method is beneficial.

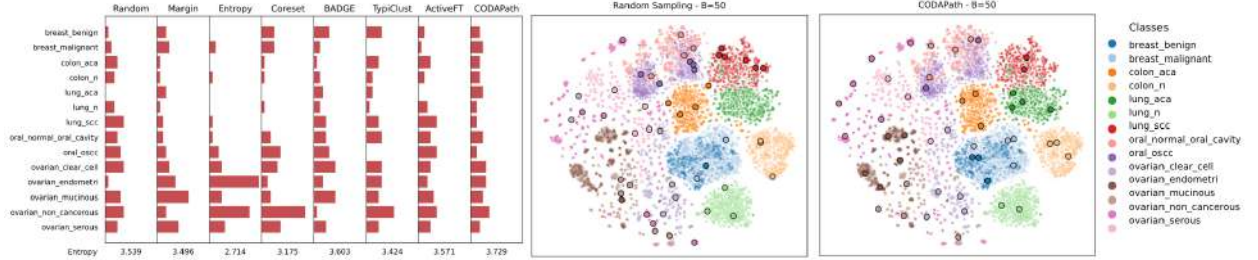


Fig. 3. Qualitative analysis of sampling coverage at budget $B = 50$. **Left:** Class distribution of the initial batch for each active learning method. A higher entropy indicates a more balanced class representation. **Middle and Right:** Feature space visualization for Random Sampling and CODAPPath. Selected samples are highlighted with black boundaries. Unlike random selection, CODAPPath ensures comprehensive spatial coverage across all distinct class clusters.

under severe annotation constraints, which simulate the prohibitive annotation bottlenecks frequently encountered in real-world clinical deployments.

Superiority in the Cold-Start Regime. Our CODAPPath demonstrates a strong advantage in the most critical extreme low-budget phases ($B = 50$ and $B = 100$). Specifically, at the initial step ($B = 50$), CODAPPath achieves notable accuracy gains over the second-best performing baselines, yielding margins of +2.97% on PathMNIST, +9.84% on HistoSet-5x14, and +6.27% on SkinTissue. While cold-start baselines like TypiClust [3] and ActiveFT [22] are also effective in these early stages, CODAPPath leverages Med-VLM semantic priors to establish a more informative and resilient decision boundary from step zero. Furthermore, in specific later configurations where CODAPPath shifts to the second or third rank (e.g., at $B = 100$ on PathMNIST or $B = 150$ on HistoSet-5x14), it remains competitive. The performance gap between CODAPPath and the first-ranked method is remarkably narrow, bounded between 0.3% and 1.5% (e.g., a mere 0.89% margin at $B = 100$ on PathMNIST, and 0.50% at $B = 150$ on HistoSet-5x14). This indicates that even when not holding the absolute lead, CODAPPath avoids performance degradation and maintains stability.

Vulnerability of Traditional Methods. The results validate the vulnerability of traditional AL methods to the cold-start problem. At $B = 50$, strategies like Entropy [10] and Coreset [14] suffer catastrophic failures, performing significantly worse than naive Random sampling (e.g., Entropy yields a mere 49.05% on HistoSet-5x14). Because the initially untrained model produces highly uncalibrated and noisy probabilities, hallucinated uncertainty estimations actively mislead the acquisition function. As the query budget increases ($B \geq 200$) and the model becomes reasonably calibrated, iterative methods like Margin [11] predictably recover and converge toward state-of-the-art performance. However, CODAPPath’s robust zero-shot initialization ensures it remains highly competitive across all budget levels without wasting the early annotations.

4.2. Qualitative Evaluation

To understand why CODAPPath performs strongly in the cold-start phase, we analyze the quality of the initial batch ($B = 50$) selected by each method. Fig. 3 (Left) shows the distribution of sampled classes and their corresponding Shannon entropy. Traditional active learning methods frequently sample from a narrow subset of classes, leading to severe class imbalance. For example, Entropy [10] and Coreset [14] almost entirely ignore several minority classes. Our CODAPPath

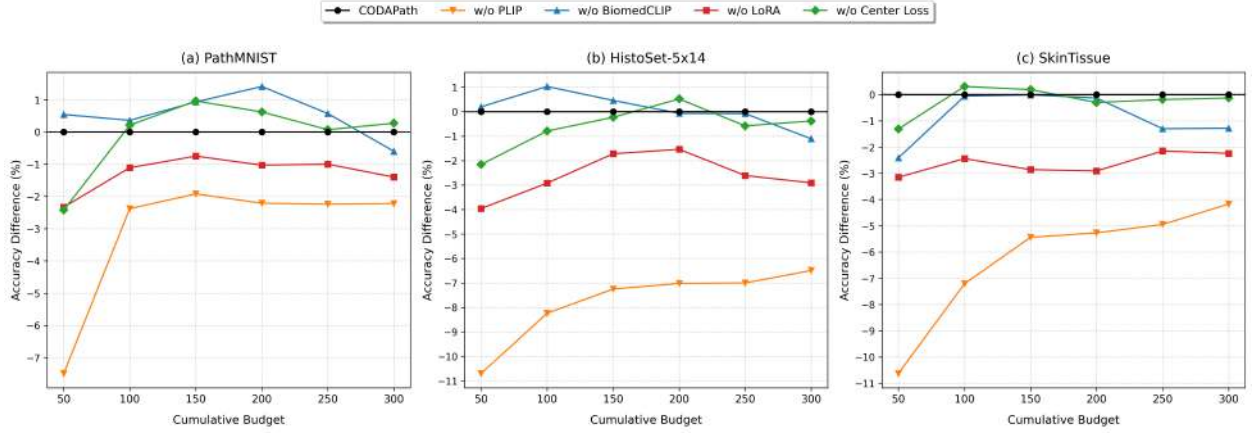


Fig. 4. Ablation study on core components. The figure illustrates the accuracy degradation when removing specific modules from the full CODAPath framework. A negative difference indicates a performance drop, confirming the essential contribution of the ablated component.

uses its zero-shot textual priors to acquire a highly balanced batch without needing actual labels, achieving the highest entropy score. This balanced start prevents the model from overfitting to majority classes during the critical early stages of fine-tuning.

Fig. 3 (middle and right) provides a visual explanation of this sampling behavior in the feature space. When using Random sampling, the selected points (marked with black boundaries) tend to clump heavily in high-density regions. This leaves several distinct morphological clusters completely empty (e.g. *lung_aca* or *ovarian_endometri* classes), meaning the model has zero data to learn those specific tissue patterns. Our CODAPath actively forces the selection of points across the entire feature manifold. By placing anchor points in every distinct cluster, CODAPath encourages comprehensive coverage of the dataset’s structural variety. This direct connection between balanced sampling and broad spatial coverage likely contributes to the difference in accuracy observed at the $B = 50$ budget level in Table 1.

Furthermore, this balanced initialization directly synergizes with our fine-tuning objective. Because CODAPath successfully anchors samples in every distinct morphological cluster, Center Loss (L_{center}) can immediately begin pulling local feature neighborhoods toward well-defined class prototypes from the very first epoch. In contrast, when baseline methods like Random or Entropy fail to sample a minority class entirely, standard cross-entropy cannot establish a valid decision boundary for that class. Thus, the joint uncertainty-coverage acquisition acts as a necessary prerequisite for the subsequent contrastive optimization, improving downstream optimization even when $B = 50$.

4.3. Ablation Study

To validate the contribution of each proposed module, we conduct a comprehensive ablation study by isolating core components of our CODAPath.

Specifically, we evaluate the framework’s performance without PLIP (*w/o PLIP*), without BiomedCLIP (*w/o BiomedCLIP*), without Low-Rank Adaptation (*w/o LoRA*), and without the auxiliary contrastive objective (*w/o Center Loss*). The empirical accuracy degradation relative to the full framework across varying budgets is visualized in Fig. 4.

Impact of Dual-Foundation Zero-Shot Priors.

The empirical results demonstrate that domain-specific vision-language alignment is the primary driver for solving the cold-start problem. Removing PLIP leads to substantial performance degradation in the extreme low-budget regime. For instance, at $B = 50$, the accuracy plummets by 10.70% on HistoSet-5x14 and 10.63% on SkinTissue. Without these powerful semantic priors, the acquisition function essentially reverts to blind guessing at step zero. Furthermore, while PLIP dominates the feature representation, removing BiomedCLIP also results in noticeable performance drops on specific datasets (e.g., dropping by 2.41% on SkinTissue at $B = 50$). This supports the motivation for the dual-foundation design: ensembling multiple Med-VLMs broadens the semantic coverage and mitigates the domain bias inherent in relying on a single foundation model.

Necessity of Parameter-Efficient Fine-Tuning.

Fine-tuning a high-capacity dual-encoder architecture with a minimal batch of 50 or 100 samples can increase the risk of overfitting. When LoRA is removed and the entire network is subjected to full fine-tuning, performance degrades consistently across all datasets. Most notably, at $B = 50$, *w/o LoRA* suffers a 3.96% drop on HistoSet-5x14 and a 3.15% drop on SkinTissue. By constraining the optimization space to low-rank matrices, LoRA successfully preserves the rich pre-trained representations while efficiently adapting the model to the target diagnostic task.

Effectiveness of Center Loss Optimization. Under extreme data scarcity, standard cross-entropy fails to explicitly structure the geometric distribution of the

feature space. The ablation of Center Loss (*w/o Center Loss*) results in a consistent accuracy drop across all budgets, particularly evident at the initial step $B = 50$ (-2.42% on PathMNIST and -2.16% on HistoSet-5x14). This validates our geometric intuition: by actively pulling same-class samples toward learnable shared prototypes, Center Loss provides the essential regularization needed to enforce compact intra-class clusters and maximize the discriminative power of the minimally constrained labeled pool.

5. Conclusion

In this paper, we present CODAPath, a novel Active Learning framework for extreme low-budget pathology image classification. CODAPath eliminates reliance on random initialization by leveraging dual-foundation Med-VLMs (PLIP and BiomedCLIP) as zero-shot priors, enabling robust sample utility estimation before any annotation occurs. A joint greedy acquisition strategy simultaneously balances predictive uncertainty and feature-space coverage, while Center Loss during fine-tuning enforces compact intra-class clusters and well-separated inter-class margins under severe data scarcity. Experiments across three public pathology datasets demonstrate that CODAPath performs competitively and often outperforms state-of-the-art AL baselines under extreme low-budget constraints. At just 50 annotated samples, CODAPath achieves substantial accuracy gains over the best-performing competitors, supporting the benefit of integrating medical semantic priors from step zero. In future work, we will explore extending CODAPath to open-set scenarios, where unlabeled pools may contain out-of-distribution samples irrelevant to the target task.

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Appendix

A. Med-VLM Configurations

A.1. Morphological Class Descriptions

Relying strictly on raw class names provides insufficient semantic context for the vision-language models to establish robust zero-shot decision boundaries. To address this, we define detailed, clinically accurate morphological descriptions for all 39 categories across the three datasets. Due to space constraints, we provide a representative subset of these descriptions below, illustrating the contrast between the raw class labels and the enriched semantic priors. The complete mapping is available in our supplementary source code.

Dataset: PathMNIST

normal_colon_mucosa: benign, healthy normal colon mucosa and colonic glands.

colorectal_adenocarcinoma: colorectal adenocarcinoma epithelial cells forming malignant glands.

Dataset: HistoSet-5x14

lung_n: benign, healthy and normal lung tissue without malignancy.

lung_aca: lung adenocarcinoma, consisting of malignant glandular epithelial cells.

oral_oscc: oral squamous cell carcinoma, showing invasive malignant squamous cells with keratin pearls.

Dataset: SkinTissue

tumor_skin_epithelial_bcc: basal cell carcinoma, displaying nests of atypical basaloid cells with characteristic peripheral palisading and stromal retraction artifact.

nontumor_skin_elastosis: solar elastosis, showing basophilic degeneration and accumulation of thick, clumped abnormal elastic fibers in the dermis due to sun damage.

nontumor_skin_dermis: normal dermal connective tissue, consisting of dense interwoven collagen bundles and scattered fibroblasts.

A.2. Prompt Templates

To bridge the semantic gap between visual features and clinical text, we project domain-specific knowledge into the Med-VLM text encoders using $K = 5$ unified prompt templates. The placeholder `[CLS_DESC]` in each template is dynamically replaced by the morphological description of the respective class. The exact templates utilized in our framework are:

- “Photomicrograph showing `[CLS_DESC]`.”
- “Histopathology of `[CLS_DESC]`.”
- “H&E stained microscopic image of `[CLS_DESC]`.”
- “Pathological tissue section demonstrating `[CLS_DESC]`.”
- “Cellular morphology characteristic of `[CLS_DESC]`.”

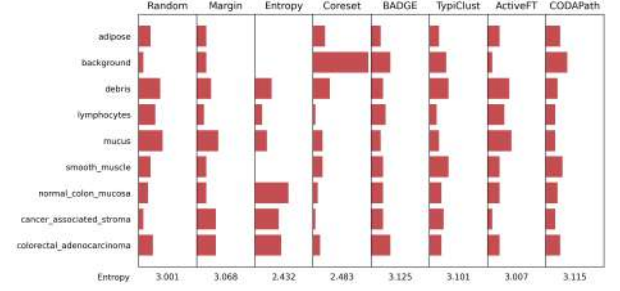
B. Implementation Details

Table 2. Detailed training configurations and hyperparameters.

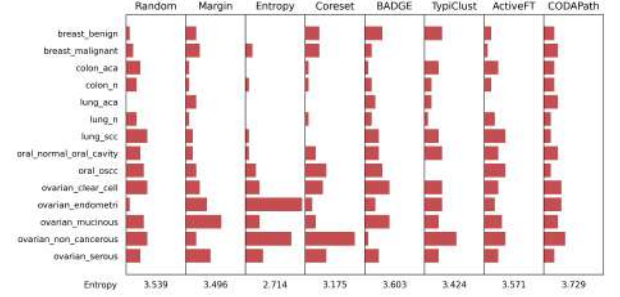
Hyperparameter	Value
Hardware & Setup	
Hardware	1 × Tesla T4 (16GB)
Framework	PyTorch
Mixed Precision	FP16
Dual-Encoder Architecture	
PLIP Backbone	vinid/plip
BiomedCLIP Backbone	microsoft/BiomedCLIP-PubMedBERT_256-vit_base_patch16_224
Backbone Status	Frozen (Feature Extraction)
LoRA	$r = 8$, $\alpha_{\text{LoRA}} = 16$, Dropout=0.1
PLIP Targets	q_proj, v_proj
BiomedCLIP Targets	qkv, in_proj, out_proj
Shared Projection Head	Linear(256) → BatchNorm1D → ReLU → Dropout(0.4)
Active Learning & Optimization	
Temperature (τ)	0.05 (Eq. 3)
AL Trade-off (α)	0.5 (Eq. 8)
Main Opt.	AdamW (1×10^{-4})
(LoRA+Head)	
Weight Decay	1×10^{-4}
Center Opt.	SGD (LR=0.5)
Center Loss (λ)	0.05 (Eq. 9)
Epochs per round	25
Batch Size (N)	min(32, Budget B)

B.1. Dataset Partitioning

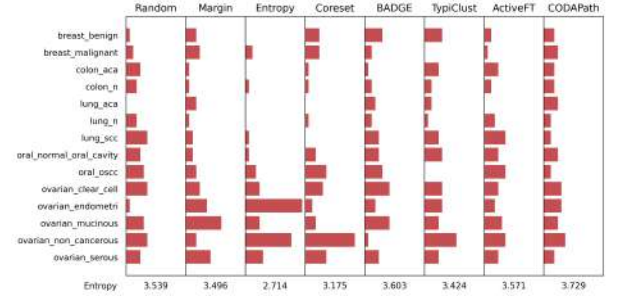
To rigorously evaluate the model’s performance on the cold-start active learning task, we simulate a standard oracle-based annotation scenario. For the HistoSet-5x14 and SkinTissue datasets, the entire corpus is randomly partitioned into an 80% unlabeled training



(a) PathMNIST



(b) HistoSet-5x14



(c) SkinTissue

Fig. 5. **Class distribution of the initial batch.** Visualization across three evaluation datasets at budget $B = 50$.

pool (from which the active learning acquisition function queries samples) and a 20% hold-out test set for evaluation. For the PathMNIST dataset, we adhere to its official distribution splits; however, to maximize the available unlabeled pool, we merge the official *train* and *validation* sets to form the training pool, while keeping the official *test* set strictly isolated for final evaluation.

B.2. Training Configurations

To ensure full reproducibility of our experiments, Table 2 outlines the complete set of hyperparameters, architectural configurations, and hardware specifications used to train our CODAPath framework and other active learning methods.

C. Additional Comparisons

Expanding on Section 4.2, we provide the class distributions and corresponding Shannon entropy of the initial batch ($B = 50$) for PathMNIST (Fig. 5a), HistoSet-5x14 (Fig. 5b), and SkinTissue (Fig. 5c).

CODAPath consistently selects a diverse initial batch across datasets, achieving the highest label entropy on HistoSet-5x14 (3.729) and SkinTissue (3.501). On PathMNIST, CODAPath remains competitive with BADGE, obtaining an entropy of 3.115 compared with 3.125. These results suggest that our domain-agnostic acquisition strategy produces a more balanced starting pool and may help explain the stronger downstream performance observed across datasets.

By contrast, competing AL methods exhibit greater imbalance during the initial acquisition stage. On PathMNIST, Entropy over-selects `normal_colon_mucosa` while failing to sample `adipose`, resulting in a lower entropy of 2.432. Likewise, Coreset (entropy 2.483) allocates a disproportionate portion of the budget to non-diagnostic background patches. On the 16-class SkinTissue dataset, Entropy also fails to capture categories such as `tumor_skin_epithelial_bcc`.

By utilizing Med-VLM priors instead of uncalibrated visual features, CODAPath selects samples that better represent the target distribution. This more diverse initialization can provide a stronger foundation for Center Loss to form class prototypes and reduce the risk of majority-class overfitting.